

IS 302-2-209 (1994)

(Reaffirmed 2015)

(Reaffirmed 2020)

भारतीय मानक

घरेलू और सम्बद्ध विद्युत साधनों की सुरक्षा

भाग 2 विशेष अपेक्षाएं

अनुभाग 209 कम गति की खाद्य-प्रेषण मशीन

Indian Standard

**SAFETY OF HOUSEHOLD AND SIMILAR
ELECTRICAL APPLIANCES**

PART 2 PARTICULAR REQUIREMENTS

Section 209 Low Speed Food Grinding Machines

UDC 64.06-83 : 621.924-83 : 614.82

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BUREAU OF INDIAN STANDARDS

MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

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Price Group 4

AMENDMENT NO. 1 JULY 2000
TO
IS 302-2-209(1994) SAFETY OF HOUSEHOLD AND
SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 209 Portable Low Speed Food Grinding Machines

(Page 1, clause 5.1) — Substitute '1 000 watts' for '1 000 volts'

(ETD 32)

Reprography Unit, BIS, New Delhi, India

AMENDMENT NO. 2 SEPTEMBER 2001
TO
IS 302-2-209(1994) SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL
APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

SECTION 209 Low Speed Food Grinding Machines

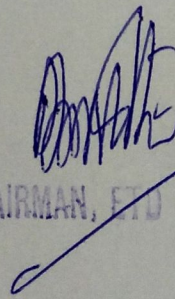
(Page 6, Clause 101.1, line 2, 3 & 4) – Substitute “a sample” for “two samples of grinding machines of the same type and rating”.

(Page 6, Clause 101.1.1, line 1) – Substitute “sample” for “both samples”

(ETD 32)

अनुमोदित
APPROVED FOR ADOPTION

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**AMENDMENT NO. 3 JANUARY 2010
TO
IS 302-2-209 (1994) SAFETY OF HOUSEHOLD AND
SIMILAR ELECTRICAL APPLIANCES**

PART 2 PARTICULAR REQUIREMENTS

Section 209 Low Speed Food Grinding Machines

(Page 1, clause 5.101, para 2) — Substitute the following for the existing matter:

‘Preferred capacities per loading shall be 5 ml, 15 ml, 50 ml, 500 ml and 1 to 3 litres.’

(ET 32)

Reprography Unit, BIS, New Delhi, India

Electrical Appliances Sectional Committee, ET 32

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrotechnical Division Council.

This Indian Standard covers the general and safety requirements of low speed food grinding machine. This standard does not cover the performance requirements, which are covered under a separate standard IS 7603 : 1975.

This standard is to be read in conjunction with IS 302-1 (1979) 'Safety of household and similar electrical appliances : Part 1 General requirements'. For the sake of convenience, the clauses of this standard correspond to those of IS 302-1 (1979) Instead of reproducing full text of each clause, clauses of IS 302-1 (1979) which are applicable (which means that relevant provisions of the clause apply) or not applicable and the sub-clauses or portions thereof which are not applicable are indicated as under:

- a) In case of a clause where it is applicable or not applicable, the wording used is 'This clause of IS 302-1 (1979) is applicable/not applicable'.
- b) In case of a sub-clause or part thereof 'Not applicable'.

Wherever a sub-clause of IS 302-1 (1979) is to be replaced by a new text, it has been indicated as under:

Replacement — followed by the next text

Any addition to the existing provisions of a sub-clause of IS 302-1 (1979) has been indicated as under:

'Addition — followed by the text of the additional matter'.

Clauses/tables which are additional to those of IS 302-1 (1979) are numbered starting from **101** and additional sub-clauses are numbered with the main clause number followed by **101, 102**, etc, for example, **7.101**.

Should, however, any deviation exist between IS 302-1 (1979) and this standard, the provisions of the latter shall apply.

While preparing this standard, assistance has been derived from the following publication issued by the International Electrotechnical Commission:

IEG Pub 335-1 (1976) Safety of household and similar electrical appliances: Part 1 General requirements

The Indian Standards which are necessary adjuncts to this standard are given in Annex AA.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 209 Low Speed Food Grinding Machines

1 SCOPE

This clause of IS 302-1 (1979) is applicable except as follows:

1.1 Replacement

This standard covers general and safety requirements for electrically motor operated low speed food grinding machines intended for domestic and similar use, their rated voltage not exceeding 250 V ac single phase or dc or 415 volt three phase ac.

1.2 Replacement

This standard does not apply to:

- electric kitchen machines which is covered under a separate standard IS 302-2-14 (1994).

For appliances intended to be used in vehicles or on board ships or aircraft, additional requirements may be necessary.

2 TERMINOLOGY

2.1 This clause of IS 302-1 (1979) is applicable except as follows:

Additional Subclauses :

2.101 Grinder

A portable appliance which by operation of low speed abrasive friction surfaces or by compression surfaces is intended primarily for pulverizing or powdering dry foodstuffs, either raw or roasted, such as cereals, grains, coffee seeds, etc, and/or converting them in the presence of water or oil, into forms of slurries or pulps.

2.102 Speed

The revolutions of the part of the appliance, where actual change of the foodstuff takes place.

2.103 Abrasive Friction Surface

Any surface that has a relative motion and in close contact with another stationary surface, having a high coefficient of friction.

2.104 Hopper

Container part of the grinder where food is

inserted and finds its way to the grinding mechanism during the operation.

2.103 Spout

That part of the grinder where the ground material flows out of the unit.

3 GENERAL REQUIREMENT

This clause of IS 302-1 (1979) is applicable except as follows:

3.1 Para 2 not applicable.

4 GENERAL NOTES ON TESTS

This clause of IS 302-1 (1979) is applicable except as follows:

4.8, 4.10, 4.13 and 4.14 not applicable.

5 RATING

This clause of IS 302-1 (1979) is applicable except as follows:

5.1 Replacement

The rated input shall not exceed 1 000 volts.

Additional Subclauses:

5.101 Rated Capacity

The rated capacity shall be the maximum specified quantity handled by the unit for each complete cycle of operation. This may be specified as the hopper capacity.

Preferred capacities per loading shall be 50, 75, 100, 300, 500 ml and 1 litre.

5.102 Rated Duty Period

The duration to which the grinder is designed to work continuously under rated load and under conditions of adequate heat discharge, without excessive heating up shall be not less than 15 minutes or the time taken for grinding the maximum specified quantity of the most adverse material whichever is greater.

5.103 Rated Frequency

Where the grinder is designed to work purely on ac circuit, the rated frequency shall be 50 Hz.

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5.104 Rated Speed

The rated speed of the grinder shall not exceed 500 rev/min and the surface velocity shall not exceed 300 m/min.

6 CLASSIFICATION

This clause of IS 302-1 (1979) is applicable except as follows.

7 MARKING

This clause of IS 302-1 (1979) is applicable except as follows:

7.101 Marking of Control Switch

The 'on' or 'off' or both positions of the control shall be clearly marked. If any speed control device is provided, the various positions shall be clearly and indelibly marked.

7.102 Marking of Grinding Control

Various positions of the grinding adjustment lever or control shall be clearly and indelibly marked to indicate the consistency expected in each position. The positions shall have a direct relation to the ground product size in terms of millimetre and microns sieve designations in IS 460 : 1962.

7.103 Marking of Earth Terminal

The symbol shall be marked permanently on or adjacent to the earth terminal, if provided. When coloured wires are used, green colour shall be used for earthing wire.

7.104 Instructions for Use

Each grinder shall be provided with necessary instructions for use which will include among other informations the following:

a) Precautions

- 1) While positioning of the grinder, and
- 2) Before switching on the machine.

b) Warning

- 1) About the parts of the grinder which shall not be brought into contact with liquids, and
- 2) About the parts which may prove dangerous when in motion.

c) Instruction

- 1) For assembling and dismantling the hopper and other associated parts, cleaning procedure, and servicing;
- 2) For type of supply to which the grinder is to be connected and instructions for electrical connections; and

- 3) For keeping the grinder cleaned and dried when not in use.

- d) Directions during abnormal operations or when the grinder is locked up.

- e) Nominal maximum duration of working the grinder at various settings.

- f) Guide for loading, indicating maximum quantity of at least three types of heavy and three types of light grains, ideal settings for desired grinding and in case of multispeed units, the ideal speed position of the control.

- g) Availability of spares and service.

8 PROTECTION AGAINST ELECTRIC SHOCK

This clause of IS 302-1 (1979) is applicable. **8.2, 8.3** and **8.7** not applicable.

9 STARTING OF MOTOR-OPERATING APPLIANCES

This clause of IS 302-1 (1979) is applicable.

10 INPUT AND CURRENT

This clause of Part 1 is applicable except as follows:

10.1 Addition

For multiple units the total input may be determined by measuring the input of each and separately.

The rated input is the sum of the rated inputs of all the individual elements in the appliances which can be on at one time; where there are several such combinations possible, that giving the highest input is used in determining the rated input.

11 TEMPERATURE RISE

This clause of IS 302-1 (1979) is applicable except as follows:

11.7 Addition

The grinder shall be operated under conditions of normal loading, while connected to a supply circuit at maximum rated voltage. To obtain normal loading conditions the hopper shall be filled to 70 percent of its rated capacity with a lubricating oil or mixture of oils whose flash point exceeds 650°C and of such viscosity as to give consistently the maximum deflection of the wattmeters and shall be operated at the setting of the grinder control which gives the lowest grinding size. The test shall be conducted over one complete duty period. Initially the spout is kept closed. After the machine is started, the spout is

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kept open. The hopper is continuously kept filled to the same capacity with an external oil supply and at room temperature.

The temperature of any material or insulation which may be exposed to excessive temperature during the operation shall be measured, by change of resistance or thermocouple method. The appliance shall be rejected if the temperature-rise so determined over an ambient temperature not exceeding 40°C, exceeds the temperature permissible according to Table 1 of IS 302-1 (1979).

Thermal cutouts and overload protectors, if any, shall not operate during the test.

Additional Subclause

11.101 Temperature Withstand Test (for Plastic and Glass Hoppers Only)

Glass hoppers will be tested without assembling them on grinders. The emit port is closed by artificial means, and held in the position of normal use. Boiling water is poured into this hopper to fill it to its brim over a period of half a minute.

After this test the hopper is emptied and brought back to room temperature within 5 minutes by forced air-cooling.

The above test shall be repeated 5 times. At the end of the test the hopper shall not show any sign of cracks.

12 OPERATION UNDER OVERLOAD CONDITIONS OF APPLIANCES WITH HEATING ELEMENTS

This clause of IS 302-1 (1979) is applicable.

13 ELECTRICAL INSULATION AND LEAKAGE CURRENT AT OPERATING TEMPERATURE

This clause of IS 302-1 (1979) is applicable.

14 RADIO AND TELEVISION INTERFERENCE SUPPRESSION

This clause of IS 302-1 (1979) is applicable.

15 MOISTURE RESISTANCE

This clause of IS 302-1 (1979) is applicable except as follows:

15.3 Addition

For wet grinder the output or the spout shall be temporarily blocked for the purpose of this test; the hopper shall be assembled and water shall be poured slowly over a period of one minute into the grinder through the hopper, to fill it entirely, taking care not to spill water over the sides or on the grinder. The grinder shall be allowed to stand in this condition for a period of 15 minutes. At

the end of this period the grinder shall be examined for any visible signs of leakage or any drop in the level of water in the hopper. If there is drop in the level measured, a quantity of water is again poured into the hopper slowly to bring the level to original. The entire quantity of water is now drained out of the grinder and then measured. Leakage of water in this test shall not be more than 2 percent of the total quantity of water content.

The grinder is once again filled as above and 20 percent more water (total quantity) is made to spill over the hopper over a period of 1 minute.

The machine shall then withstand the tests specified in **16.4**.

If it is not possible to place the whole appliance in the humidity cabinet, parts containing electrical components are tested separately, taking into account the conditions which occur in the appliance.

16 INSULATION RESISTANCE AND ELECTRIC SHOCK STRENGTH (AFTER HUMIDITY TREATMENT)

This clause of IS 302-1 (1979) is applicable except as follows:

16.3 Not applicable.

17 OVERLOAD PROTECTION

This clause of IS 302-1 (1979) is applicable.

18 ENDURANCE

(Under consideration).

19 ABNORMAL OPERATION

This clause of IS 302-1 (1979) is applicable except as follows:

19.1 Replacement

Appliances shall be so designed that the risk of fire, mechanical damage impairing safety or the protection against electric shock as a result of abnormal or careless operation is obviated as far as is practicable.

Compliance is checked by the tests of **19.2** to **19.4**.

Fuses, thermal cut-outs, overcurrent releases or the like incorporated into the appliance, may be used to provide the necessary protection.

If more than one of the tests are applicable for the same appliance, these tests are made consecutively.

Provisions of **19.11** apply to all appliances.

19.2 Replacement

Appliances, without water load and with all doors or lids closed, are operated under the conditions

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specified in clause 11, the supply voltage, determined prior to the test, being that required to provide an input of 0.85 times the rated input under conditions of adequate heat discharge when steady conditions have been established. This voltage is maintained throughout the test.

If a non-self-resetting thermal cut-out operates, or if a current is otherwise interrupted before steady conditions are established, the operating period is considered to be ended and the test of 19.7 is not made.

If interruption of the current does not occur, the steam cooker is switched off as soon as steady conditions are established, and is allowed to cool down to approximately room temperature. The appliance is then subjected to the test of 19.3.

20 STABILITY AND MECHANICAL HAZARDS

This clause of IS 302-1 (1979) is applicable except as follows:

20.1 Addition

Covers, lids and accessories are placed in the most unfavourable positions.

For the purpose of this test any spillage of liquid is ignored

20.2 Not applicable.

21 MECHANICAL STRENGTH

This clause of IS 302-1 (1979) is applicable.

21.2 to **21.4** not applicable.

Additional Subclause

21.101 Strength of Assembly

The hopper, if detachable, shall be assembled and mounted on grinder assembly for six times under the following controlled conditions:

- a) Press-fit joints using a force of 245 N, and
- b) Screws and screwed-on joints, using a screwing torque of 2.45 Nm.

There shall be no chipping, cracking or visible dents or scratches on the mating surfaces.

22 CONSTRUCTION

This clause of IS 302-1 (1979) is applicable except as follows:

22.1 Replacement

Appliances shall be of Class I, II or III construction.

Compliance is checked by inspection and by the relevant tests.

Additional Sub-clause

22.101 Appliances shall be protected in such a manner that moisture and grease will not collect in such a way as to affect creepage distance and clearance values.

Compliance is checked by inspection.

22.102 Thermal cut-outs shall be of the non-self resetting trip-free type.

Compliance is checked by inspection and manual test.

22.103 For fixed appliances and appliances which are permanently connected to fixed wiring, it shall be possible to gain access to components for servicing without moving the appliance, for example heating elements and controls

22.104 The heating element or elements shall be so designed and located that the heating conditions required for cooking can be obtained and local overheating or burning of foodstuffs is avoided. The elements shall be easily accessible for inspection, repair and replacement. It shall not get displaced when the steam cooker is lifted by its carrying handle, if any, or for a broken element to fall out of the steam cooker. Compliance is checked by inspection and manual test.

22.105 A thermostat shall be provided to control the temperature within the steam cooker.

22.106 Switches and controls shall be so located and protected that these are not subjected to mechanical injury, spillage from cooking or collection of grease. Compliance is checked by inspection and by test of 15.3.

22.107 The inner container of the Grinding machine shall be made of either stone or any other suitable hard material of sufficient thickness so as to make it withstand normal wear and tear.

22.108 Materials used in the grinder shall be non-flammable and corrosion resistant.

22.109 Material of the grinding wheel shall be such that it shall not react with the foodstuff and shall not contaminate with the same and the grinding wheel shall not be appreciably ground by the food material during the test specified in this standard.

22.110 Parts which would come in contact with foodstuff shall be made of cast iron, cast aluminium, sheet metal or high impact heat resistance plastic of adequate strength in case of dry grinding and for wet grinding the same shall be made of stainless steel or aluminium or plastic with high impact heat resisting properties.

22.111 All parts required to be cleaned and kept in a hygienic condition shall be readily accessible without the use of special tools.

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22.112 All castings and other materials required to be handled shall be smooth, round edged end free from blow holes, pits, foreign matter and surface imperfections.

22.113 Machined and formed parts shall be made to ensure complete interchangeability and parts subject to wear shall be easily replaceable.

22.114 A lid shall be provided to the hopper to avoid spillage or splutter of grains during the grinding.

22.115 Means shall be provided to adjust the ground grain size or slurry consistency without the use of tools. This adjustment shall stay firm for each setting, and shall not work loose due to vibrations or heating.

22.116 Means shall be provided to inspect the ground food, at the outlet spout without any necessity for stopping the machine.

22.117 Means shall be provided for easy collection of ground products.

22.118 Shafts of the grinding wheels and other associated parts shall be suitably constructed to ensure against the ingress of liquids, oil, fats or particles of foodgrains, so as not to impair the performance of the machine.

22.119 Ventilating Openings

Openings for ventilation of the motor shall be properly screened, generally to prevent entry of water, dirt or vermin.

22.120 Electric Motor

The motor shall conform to the requirements of IS 996 : 1979 and IS 325 : 1978. It may be provided with a facility to operate at more than one speed.

22.121 Controls

Switches and controls shall have stable positions of operation at various speeds.

22.122 Hopper

The hopper shall have a means for shutting off or opening the flow of grains into the grinder mechanism. The hopper may be designed to be easily removable from the machine, and shall be free from cracks, pits and crevices. The inner sides shall be smooth and shall not have corners or niches so that cleaning is facilitated. The fixing arrangement of the hopper shall be adequately strong to stand repeated operations. It may be fitted with handling grips. When the hopper, if detachable, is fitted on the grinder, it shall be leak-proof.

22.123 Speed of Grinding Surface

The speed of the grinding surface shall not exceed 300 m/min on full load condition on the highest speed position of switch nor shall be less than 150 m/min on the lowest position of the switch.

NOTE — The speed of the surface is measured where it offers the maximum velocity, even if such a surface does not actually perform grinding operation, but constitute a part of the grinding surface.

22.124 The design shall ensure that no accidental bodily contact is made with grinder or its mechanism, such as intake-screws while they are in the operation.

22.125 The design shall ensure that lubrications and abrasive materials do not contaminate the food while it is being processed. Grinding surfaces shall be corrosion and rust resisting.

22.126 Control switches shall be capable of breaking the stalled motor current at the maximum rated voltage 10 times without failure. This test shall be carried out on dc unless the machine is marked exclusively for ac.

22.127 Under normal load, at each setting the speed when checked shall not vary by more than ± 10 percent of the marked speed at that setting. Compliance is checked by inspection.

23 INTERNAL WIRINGS

This clause of IS 302-1 (1979) is applicable.

23.2 Replacement

The wiring and all the components used in the grinding machine shall be constructed and fixed in such a manner that these shall be able to withstand continuously the temperature to which these are exposed when the steam cooker is operated at its maximum temperature. All the wiring and components shall be suitably protected from spillage or overflow from the inner container of the grinding machine.

24 COMPONENTS

This clause of IS 302-1 (1979) is applicable except as follows:

24.9 and **24.10** Not applicable.

24.11 Replacement

Thermal cut-outs shall provide all phase disconnection from the supply.

Compliance is checked by inspection.

25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CABLES AND CORDS

This clause of IS 302-1 (1979) is applicable except as follows:

The terminals for permanent connection to fixed wiring may also be suitable for the type X attachment of a power supply cord. In this case, the instruction sheet shall give full particulars of the power supply cord.

Compliance is checked by inspection.

25.4 Addition

Type X attachment shall be used for appliances delivered without a cord.

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25.6 Replacement

Instead of the types of power supply cord specified, the following applies:

Power supply cords shall be polyvinyl insulated and sheathed flexible cord or oil resistant sheathed flexible cable not lighter than ordinary polychloroprene or other equivalent synthetic elastomer-sheathed cord [see IS 9968 (Part 1) : 1988 and IS 694 : 1990]. The length of the supply cord shall not be less than 2.0 m if supplied with a plug or 2.05 if supplied without plug.

25.6 Addition

The length of the supply cord measured between the entry point into the appliance and the entry point into the plug and cord guard included, shall be not less than 2.0 m if supplied with plug, or shall be not less than 2.05 if supplied without plug.

25.10 Not applicable.

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of IS 302-1 (1979) is applicable.

27 PROVISION FOR EARTHING

This clause of IS 302-1 (1979) is applicable except as follows:

27.2 Addition

Small fixed exposed metal parts, for example name plates and the like, are not required to be in electrical contact with the terminal.

28 SCREWS AND CONNECTIONS

This clause of IS 302-1 (1979) is applicable.

30 RESISTANCE TO HEAT, FIRE AND TRACKING

This clause of IS 302-1 (1979) is applicable.

31 RESISTANCE TO RUSTING

This clause of IS 302-1 (1979) is applicable.

32 RADIATION HAZARDS

This clause of IS 302-1 (1979) is applicable.

101 TESTS

101.1 Type Test

The tests specified in Table 1-1 shall constitute the type tests and shall be carried out on two samples of Grinding Machines of the same type and rating selected preferably at random from a regular production lot. Before commencement of the tests, the steam cookers shall be visually examined and inspected for obvious visual defects in respect of components, part and their assembly,

construction, mechanical hazards, markings, provision of suitable terminals for supply connections, earthing and the effectiveness of screws and connections. The external surface finish shall be even and free from finishing defects.

NOTE — See also 4.2.

101.1.1 Criteria of Acceptance

Both samples shall successfully pass all the type tests for proving conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority at its discretion, may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test(s) in which failure(s) had occurred. No failure should be permitted in the repeat test(s).

Table 1-1 Schedule of Type Tests
(Clause 101.1)

Sl No.	Clause Reference
i) Protection against electric shock	8
ii) Input	10
iii) Temperature rise	11
iv) Operation under overload conditions	12
v) Electrical insulation and leakage current at operating temperature	13
vi) Radio and television interference suppression	14
vii) Moisture resistance	15
viii) Insulation resistance and electric strength (after humidity treatment)	16
ix) Overload protection	17
x) Endurance (Under Consideration)	18
xi) Abnormal operation	19
xii) Stability and mechanical hazards	20
xiii) Mechanical strength	21
xiv) Construction	22
xv) Internal wiring	23
xvi) Components	24
xvii) Cord grip and cord guard	25
xviii) Terminals for external conductors	26
xix) Provision for earthing	27
xx) Screws and connections	28
xxi) Creepage distance and clearances	29
xxii) Resistance to heat time and tracking	30
xxiii) Resistance to rusting	31
xxiv) Radiation hazards	32

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101.2 Acceptance Tests

The following shall constitute the acceptance tests:

<i>Test</i>	<i>Clause Reference</i>
a) Protection against electric shock	8
b) Input	10
c) Temperature rise	11
d) Electrical insulation and leakage current at operating temperature	13
e) Moisture resistance	15
f) Insulation resistance and electric strength (after humidity treatment)	16
g) Provision for earthing	27

NOTE — For the purpose of acceptance tests, the humidity treatment shall be done for 24 hours while conducting the test for moisture resistance (15).

101.2.1 A recommended sampling procedure for acceptance tests is given in Appendix B of IS 302-1 (1979).

101.3 Routine Test

The following shall constitute the routine tests:

<i>Test</i>	<i>Clause Reference</i>
a) Protection against electric shock	8
b) High voltage	13.3.2 of IS 302-1 (1979)
c) Provision for earthing	27

ANNEX A

TABLES OF TYPE TEST

Appendix A of IS 302-1 (1979) is not applicable.

ANNEX B

(*Clause 101.2.1*)

SAMPLING PROCEDURE FOR ACCEPTANCE TESTS

Appendix B of IS 302-1 (1979) is applicable.

ANNEX C

ELECTRONIC CIRCUITS

Appendix C of IS 302-1 (1979) is applicable if electronic circuits are used.

ANNEX D

MEASUREMENT OF TEMPERATURE WITH THERMOMETERS

Appendix D of IS 302-1 (1979) is applicable.

ANNEX E

ALTERNATIVE TESTS FOR PROTECTED MOTOR UNITS

Appendix E of IS 302-1 (1979) is applicable.

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ANNEX F

IMPACT TEST APPARATUS

Appendix F of IS 302-1 (1979) is applicable.

ANNEX G

THERMAL CONTROLS AND OVERLOAD RELEASES

Appendix G of IS 302-1 (1979) is applicable.

ANNEX H

MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES

Appendix H of IS 302-1 (1979) is applicable.

ANNEX J

TEST FOR FIRE RESISTING PROPERTIES

Appendix J of IS 302-1 (1979) is applicable.

ANNEX K

BNF JET TEST FOR DETERMINATION OF THICKNESS OF COPPER AND NICKLE PLATING

Appendix K of IS 302-1 (1979) is not applicable.

ANNEX L

APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM ON NICKEL, STEEL AND COPPER

Appendix L of IS 302-1 (1979) is not applicable.

ANNEX AA

(Foreword)

LIST OF REFERRED INDIAN STANDARDS

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
302-1 (1979)	Safety of household and similar electrical appliances : Part 1 General requirements (<i>fifth revision</i>)	996 : 1979	ing voltages up to and including 1 100 V (<i>third revision</i>) Single phase small ac motors (<i>second revision</i>) (<i>under revision</i>)
325 : 1978	Three phase induction motors (<i>fourth revision</i>)	9968 (Part 1) : 1988	Elastometer insulated cables : Part 1 For working voltages up to and including 1 100 volts (<i>first revision</i>)
694 : 1990	PVG insulated cables for work-		

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Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

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235 15 19, 235 23 15

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